Projects

Twenty projects from the portfolio, in site order. Each entry is the description on the site, followed by the written sections of that project.

---

01 Arkadium

2026 · ML & Data Analytics · Consulting · Featured

With Arkadium and the Iovine and Young Academy, this study asks whether frontier models can follow rules when the world is a game. Claude Opus, Gemini, and Grok were scored across 5 games and 115 episodes.

OVERVIEW

We asked a simple question: can frontier models actually follow rules when the world is a game? I helped design and run an evaluation of Claude Opus, Gemini, and Grok across 5 Arkadium games and 115 episodes, scoring every turn in real time as a rule violation, a legal but sub-optimal move, or a strategic one.

WHAT THE SCORING SHOWED

Models were strong at logical deduction — 94.3% competency on Ancestral Lines — and weak wherever spatial awareness mattered. Play also got worse as sessions ran longer, and 18.1% of all moves were illegal, including nearly a third of Grok's. Gameplay here is a stress test for constraint-following, not just a leaderboard.

- 94.3% competency on Ancestral Lines, where the task is logical deduction
- Spatial games were the consistent weak point
- Play degraded as a session ran longer
- 18.1% of all moves were illegal, including nearly a third of Grok's

LLM Evaluation · Research · Python · Prompt Design · Data Analysis

---

02 The Shade Gap

2025 · ML & Data Analytics · Consulting · Featured

An interactive look at Los Angeles shade inequality: canopy and heat maps, policy context including redlining, and practical responses ahead of the 2028 Olympics, including tree-spacing changes and cool-roof ideas drawn from Shade-LA and New York.

OVERVIEW

For this project we researched the current problem area, identifying key issues and historical practices like redlining which have exacerbated this issue.

WHAT THE MAPS MADE VISIBLE

To communicate the problem more effectively, we used data visualization techniques to map canopy coverage against heat levels, revealing clearly how poorer neighborhoods receive less shade and, as a byproduct, experience higher temperatures and worse health conditions.

DIRECTIONS WE EXPLORED

Following the research, we looked for solutions, drawing heavily from Shade-LA. We came up with an easy to implement change to tree spacing guidelines, as well as drawing inspiration from New York's cool-roof solution. We also researched next steps and ways to realistically implement these ideas.

Figma · Photoshop · Geospatial & Policy Research · Data Visualization · Consulting

---

03 Spotify Data Analysis

2026 · ML & Data Analytics · Consulting · Featured

A BUAD 312 consulting analysis for Universal Music Group. The work asks how release size, track duration, and artist followers are associated with streams in Hip-Hop/Rap, Pop, and Latin, then turns those models into pilot recommendations.

OVERVIEW

The project is a BUAD 312 midterm consulting analysis for Universal Music Group. The stated mission is to help UMG develop practical rollout strategies that increase digital streams. The business question is: how are release size, track duration, and artist followers associated with streams in Hip-Hop/Rap, Pop, and Latin?

The models are framed as associations, not causes. Release size is categorical (1 track, 2–6 tracks, and 7+ tracks). Duration is measured in minutes. Artist following is modeled as the log of followers and is treated as a marker of audience position, not proof that followers cause streams. Streams were observed in one snapshot, so release year compares cohorts rather than tracking the same songs over time.

DATA

The repository analyzes tracks_simplified.csv, a Spotify track table with streams, artist followers, genre, album length, release date, and duration. The presentation reports 5,870 raw tracks, 5,824 cleaned tracks, and a focus sample of 3,199 tracks in the three largest named genres, with release years from 1939 to 2021.

Hip-Hop/Rap accounts for 1,224 tracks (mean streams 49,132, median 28,460). Pop accounts for 1,633 (mean 27,075, median 14,171). Latin accounts for 342 (mean 46,098, median 26,225). Other/Unknown holds 1,996 tracks and is left out because it does not support a genre strategy. The deck flags Latin results for extra caution because that sample has only 342 tracks.

APPROACH

The analysis is written in R, using tidyverse and moderndive. It moves from descriptive comparisons to regression: basic models, parallel models that let genres differ in level, interaction models that let the rollout relationship differ by genre, and a controlled model that holds the other rollout factors and release cohort constant.

Evaluation uses p-values, R-squared, adjusted R-squared, RMSE, residual plots, and normal Q-Q plots. Coefficients are translated back into predicted percentage differences so the comparison is readable as streams, not only as log-stream slopes. Because raw streams are strongly right-skewed, the outcome is log streams. High stream and follower counts are kept, since deleting hit songs would remove observations the question is about.

WHAT THE CHARTS SHOW BEFORE THE MODELS

Duration slopes down in all three genres. Latin shows the clearest negative slope; Pop and Hip-Hop/Rap are negative but flatter. The follower relationship looks strongest in Hip-Hop/Rap and nearly flat in Pop.

Release size is grouped as 1 track, 2–6 tracks, and 7+ tracks and compared inside release periods. Singles show strong growth across those periods, especially in Pop and Hip-Hop/Rap. The 2–6 track group for Latin in 2015–2017 contains only 10 tracks.

MODEL COMPARISON

Before genre interactions, a full multiple regression is the strongest of the compared models and the lowest RMSE, and it is highly significant ($p = 3.03 \times 10^{-29}$). It still explains only a small share of log-stream variation: R^2 0.04397, adjusted R^2 0.04247, RMSE 1.348. On their own, track duration has the highest single-factor adjusted R^2 in that comparison (0.01423), release size is next (0.00567), and artist followers are lowest (0.00256) even though the followers model is statistically significant ($p = 0.00205$).

A later sensitivity check on the controlled model removes the only observation with a standardized residual beyond 3. Adjusted R^2 moves from 9.0% to 9.2%, and RMSE moves from 1.312 to 1.310. The deck says the main ranking stays stable. Residual plots still show uneven prediction error, and Q-Q tails depart from normality.

HIP-HOP/RAP

Artist followers are the factor that holds up for Hip-Hop/Rap. In the controlled comparison, an established artist at the genre's 75th follower percentile is predicted at 28,789 typical streams, against 21,273 for an emerging artist at the 25th percentile: about 35% higher. A doubling of followers is associated with about 8.1% more predicted streams per track, holding duration, release size, and year constant.

The followers model is significant on its own ($p = 3.185 \times 10^{-6}$) and after controls ($p = 1.19 \times 10^{-6}$), with adjusted R^2 0.01681 and RMSE 1.304. Release size is not significant in this genre (controlled $p = 0.9624$). Duration is significant in the simple model ($p = 0.004891$) and not after controls ($p = 0.1506$). The Hip-Hop/Rap full model has R^2 0.05916, adjusted R^2 0.0553, and RMSE 1.276.

POP

Release size is the Pop result the deck treats as the clearest predictor. With a single indexed at 100 predicted streams per track, an album track (7+ tracks) is indexed at 76.2 and an EP track (2–6 tracks) at 56.4. The single is 1.31× the predicted streams of an album track and 1.77× an EP track, holding duration, followers, and year constant.

Release size is significant in the simple model ($p = 3.38 \times 10^{-9}$) and after controls ($p = 3.39 \times 10^{-6}$), with adjusted R^2 0.02245 and RMSE 1.289. Duration and followers are not significant once controlled ($p = 0.1038$ and $p = 0.6102$). The Pop full model has R^2 0.03604, adjusted R^2 0.03308, and RMSE 1.281.

LATIN

Duration is the Latin result. Each additional minute is associated with 39.3% lower predicted streams per track. Against the 3.51-minute median, the model predicts 10.7% more streams at 3.31 minutes and 20.1% more at 3.15 minutes. Duration stays significant after controlling for release size, followers, and year (controlled $p = 8.465 \times 10^{-7}$).

Release size is also significant for Latin, but it explains less (simple $p = 0.03678$, controlled $p = 0.0329$, adjusted R^2 0.0135). Artist followers are not significant (controlled $p = 0.7162$). The Latin full model has R^2 0.1026, adjusted R^2 0.0892, and RMSE 1.533. Residuals stay within ±3 standard deviations. The deck reads the fit as good enough to set direction for a pilot, not to forecast an exact stream count.

RECOMMENDATIONS IN THE DECK

Hip-Hop/Rap: for developing artists, test campaigns, events, brand partnerships, and collaborations that build audience reach, on a limited budget, and expand only when follower growth shows up as streaming engagement.

Pop: use a single-first rollout, releasing promising tracks individually before packaging songs that are working into an EP or album. Scale only if streams and saves improve.

Latin: test 3.1–3.3 minute tracks across 10–20 upcoming releases against comparable tracks near the 3.51-minute median, keep promotion comparable, and watch completion, saves, and playlist adds.

The pilot rule in the deck is to expand a strategy only after it improves streams without weakening listener engagement or campaign returns.

LIMITS

The repository's own writeup treats the work as observational. Streams are a snapshot, so release year is a cohort, not a history of the same songs. Followers may have grown after a song succeeded. The file does not include marketing spend, playlist placement, costs, or margins, and tracks from the same artist or album may not be independent. A higher predicted stream count is not, by itself, a profit recommendation. Explanatory power stays modest even in the best models here, and the residual checks do not support precise forecasts.

R · tidyverse · moderndive · Regression · Data Visualization

---

04 Shift Website

Product Design

UX/UI and web design for SHIFT SC.

OVERVIEW

UX/UI and web design for SHIFT SC.

UX / UI · Web Design · Figma

---

05 Pneumonia Detection

2026 · ML & Data Analytics

A clinical decision-support study comparing a convolutional neural network, an RBF support vector machine, and a random forest on about 5,800 chest X-rays.

OVERVIEW

This project asks whether machine learning can reliably distinguish pneumonia from healthy lungs in chest X-rays — and whether those predictions can help clinicians screen faster.

APPROACH

I trained and compared three models on ~5,800 grayscale X-rays: a convolutional neural network, an RBF support vector machine, and a random forest.

RESULTS

Random Forest led on this data scale (82.4% test accuracy, AUC ~0.98), with the CNN close behind at 80%. The models are framed as a second-opinion tool, not a replacement for clinical judgment. Full code lives in the pneumonia_classification folder of my ML projects repo.

Python · CNN · SVM · Random Forest · Image Classification

---

06 Clothing Categorization

2026 · ML & Data Analytics

A neural network written by hand in NumPy — weights, forward pass, backpropagation, and the training loop — trained on Fashion-MNIST's ten clothing classes.

OVERVIEW

This project implements a neural network by hand — weights, forward pass, backpropagation, and training loop in NumPy — then trains it on Fashion-MNIST. The model maps 28×28 grayscale images onto ten classes (T-shirt, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, and ankle boot).

APPROACH

Data is split into train, validation, and test sets, normalized, and run through 60 epochs of minibatch gradient descent. The work sits in my ML projects repo alongside the pneumonia classifier.

Python · NumPy · Neural Networks · Fashion-MNIST · Image Classification

---

07 Polarity

Product Design

UX and UI design for a live client furniture website.

OVERVIEW

UX and UI design for a live client furniture product.

UX / UI · Figma · Client Work

---

08 Claude 9

Consulting

Claude helps founders build. Claude 9, made with Reem K and Rania K, is a go-to-market layer for what happens after the build: intelligence, deploy and host, and optimize.

OVERVIEW

Claude 9 is a go-to-market layer for Claude, made with Reem K and Rania K. Most vibe-coded apps stall after launch — 25M+ generated, then no users, no revenue, and a loop that starves the coding tools themselves. Claude is strong from idea to build and weak from launch through growth.

THE PRODUCT

Claude 9 fills that gap with three GTM features: Intelligence (ICP targeting, live traffic, competitor gaps), Deploy & Host (one-click AWS, Docker, Kubernetes), and Optimize (autonomous rewrites measured from real data). The work covers the strategy, the product UI, and the five-year prompt-to-product roadmap.

GTM Strategy · Product Strategy · UI Design · Consulting

---

09 Sabi

Consulting

Consulting for Carmen Herrera of Sabi Juices: how to scale a Latina-owned, small-batch juice company without flattening the quality, flavors, and community mission it was built on.

OVERVIEW

Consulting for Carmen Herrera, founder of Sabi Juices — a Latina-owned, community-rooted cold-pressed juice company. The brief was growth that does not flatten what the brand was built on: small-batch quality, no HPP shortcuts, Latin-inspired flavors, and a mission around brain health and comunidad. The work sits between growth strategy and founder-led brand, so scaling does not mean giving up the values that made Sabi Sabi.

Growth Strategy · Brand Strategy · Founder Advisory · Consulting

---

10 Whiskware

Consulting

Snack & Splash places a Whiskware booth inside LA84’s Summer Play Day so kids finish Swim Bowl water-safety activities and leave with character snack packs and reusable containers.

OVERVIEW

Snack & Splash puts a Whiskware booth inside LA84's annual Summer Play Day: kids complete Swim Bowl water safety activities, then take home character-themed snack packs and reusable lunch containers. The concept gives Whiskware direct exposure to 1,000–2,500 family interactions while expanding Swim Up Hill's water-safety reach. The deck covers concept, rationale, sample outreach, community purpose, and the longer vision of turning Swim Bowl into a partnership-powered event.

Sponsorship · Partnership Strategy · Deck Design · Consulting

---

11 MOCA

2024–2025 · Brand Design and Marketing

Communications work for the Museum of Contemporary Art’s Teen Night: posters, stickers, merch, and social for a citywide event of more than 500 teens.

OVERVIEW

Planned a citywide teen night of 500+ teens on the communications team. Marketed the event in person and across social media platforms by creating promotional posters, stickers, merch (hoodies and reusable bottles), and video content. Collaborated with artists and museum professionals to promote museum inclusion for underrepresented youth.

Photoshop · Instagram · TikTok · Poster Design · Event Planning

---

12 Meme Social

ML & Data Analytics

Meme Social visualizes how social-media sentiment and posting volume relate to asset prices, with interactive charts and a scenario simulator fed by LunarCrush.

OVERVIEW

Meme Social looks at how social-media sentiment and posting volume relate to asset prices — with an emphasis on separating hype-driven market psychology from fundamentals. The site visualizes that link through interactive charts and a scenario simulator, so users can explore how shifts in online sentiment and trend dynamics may influence future price movements. Sentiment and post-frequency data come from the LunarCrush API.

Data Visualization · Sentiment Analysis · HTML / CSS / JS · PHP

---

13 Gogo

2025 · Brand Design and Marketing

GOGO blends bold 1960s fashion with West African tailoring. The posters carry that into print: three colorways of the same walking-boot composition, plus a logo and campaign graphics for PresQ Studios.

OVERVIEW

GOGO blends bold 1960s fashion with the timeless artistry of West African tailoring. These posters carry that energy into print: three colorways of the same walking-boot composition, built around the GOGO mark.

CAMPAIGN

Alongside the GOGO boot, I designed a supplementary marketing plan, logo, and advertising posters to bring the brand to life.

Poster Design · Illustration · Photoshop · Branding

---

14 Estera

2025 · Brand Design and Marketing

Estera connects American customers to Europe’s slow, handcrafted fashion houses. The identity covers logo, color, campaign language, and launch materials for site, social, and pitch.

OVERVIEW

Estera bridges continents, connecting American customers to Europe's best fashion houses. I built the brand system — logo, color, campaign language, and launch materials — around slow, handcrafted European fashion. The identity covers site, social, and pitch work that makes the house feel like a bridge between cultures, not just a storefront.

Brand Identity · Logo Design · Art Direction · Campaign · E-commerce

---

15 Driving Aquatic Inclusion in Africa

2026 · Brand Design and Marketing

Two posters for Swim Up Hill’s September 22 event: a photographic aquatic version and a cream print layout covering speakers, an LA28 quick-start, and partner marks.

OVERVIEW

Two posters for Swim Up Hill's September 22 event celebrating a US/South Africa partnership and the pledge to teach one million people to swim. The series covers featured speakers — including Paralympian Jamal Hill — an LA28 quick-start guide, and partner marks from Swim Up Hill, Local Action Marketing, and South African sport institutions. One version is photographic and aquatic; the other is a clean cream layout built for print and social.

Poster Design · Typography · Layout · Campaign

---

16 Crocs x Winter X Games

Consulting

A high-fashion, high-altitude collab pitch: EVA protection, modular pads, and an extreme silhouette, so Crocs enters extreme sportswear and X Games reaches a wider audience.

OVERVIEW

People started noticing them last winter. This deck pitches Crocs x Winter X Games as a high-fashion, high-altitude collab: EVA protection, modular pads, and an extreme silhouette so wearers become the campaign. Crocs gets a lane in extreme sportswear and a higher price point. X Games gets more viewership, more women, and people who do not already watch sports. The myth is the product — athletes as figures in the sky, Crocs as the thing they wear down.

Brand Strategy · Campaign · Product Design · Consulting

---

17 Vectora

Fall 2025 · Product Design

Vectora is a GPS concept centered on the roads people share. User research with Della Smith, Joanne Chen, and Connie Sun, then a high-fidelity Figma prototype with Renae Wang, presented to Neuron founder Matt Stoner.

OVERVIEW

Vectora is a concept for a GPS built around the roads we share. I collaborated with Della Smith, Joanne Chen, and Connie Sun on user research, then built a high-fidelity Figma prototype with Renae Wang — grinded out in two days, of course. We presented the work to Matt Stoner, founder of Neuron, and came away with invaluable feedback.

User Research · Figma · Prototyping · UX · Product Design

---

18 Mattel — Tell-A-Tale

2024 · Product Design

Tell-A-Tale gives kids dice prompts and puppets they can build, with socially and emotionally complex characters. The project runs from user research and a Figma app through Fusion 360 and a business plan pitched to Mattel designers.

OVERVIEW

Tell-A-Tale is an improv storytelling experience where anyone can be anything. It consists of dice that give kids prompts for stories and puppets they can build — featuring socially and emotionally complex characters. I created the product from ideation to prototyping and a full business plan, pitching directly to Mattel designers. Work spanned user research, a Figma app, and Fusion 360.

Product Design · Photoshop · Premiere Pro · Blender · Prototyping

---

19 Current & Future Creative Resume

Brand Design and Marketing

The current resume is muted on purpose. The future one is louder and leaves the dates off. The path ran from references and a paper sketch to a type and color system in Photoshop.

OVERVIEW

The current resume is muted on purpose: restriction, a too-big head, glasses, and the feeling of having more to do than range to do it with. The future one is louder, more relaxed, and leaves the dates off.

PROCESS

Process ran references → vibe → paper sketch → black-and-white layout in Photoshop → a type and color guide → then color, copy, shadows, and the last decorative layer.

Layout · Typography · Photoshop · Art Direction

---

20 Eth-X

Product Design

Eth-X detects bias with a self-made adversarial test set, summarizes terms for popular AI products, and offers news, explainers, and a way to file a claim.

OVERVIEW

Eth-X is a browser extension and website that exposes biases in AI while also providing tools for underprivileged communities to learn about and use AI. It detects bias using a self-made adversarial test set, breaks down complex AI policies through a summarized terms and conditions page for popular AI brands, and provides interactive resources like news articles and videos to keep users informed about recent news in the field and to bridge the AI literacy gap. Users can also file claims to report misuse or request to add a new company.

Product Design · AI Literacy · Research · Web